

PREDICTION ANALYSIS REPORT

mRS 0-2 vs 3-5 Classification with Radiomics Feature Prediction

Prediction Methodology:

- 80/20 Train/Test Split (Enhanced from 75/25)
- LASSO Feature Selection for Dimensionality Reduction
- Linear SVM Classification with Cross-validation
- Radiomics Feature Prediction for Test Set
- Comprehensive Performance Evaluation

Dataset Information:

- Clinical Patients: 76 stroke patients (2020-2022)
- Radiomics Features: 73 enhanced synthetic features
- Clinical Features: 14 variables (Age, NIHSS, etc.)
- Target Variables: mRS 0-2 vs 3-5 (binary classification)

Key Innovations:

- Train Set: 80% of patients (stratified)
- Test Set: 20% of patients (stratified)
- Radiomics Feature Prediction for Test Set
- Enhanced 80/20 Split for Better Generalization
- Multi-target Prediction (4 mRS time points)
- Comprehensive Prediction Visualization

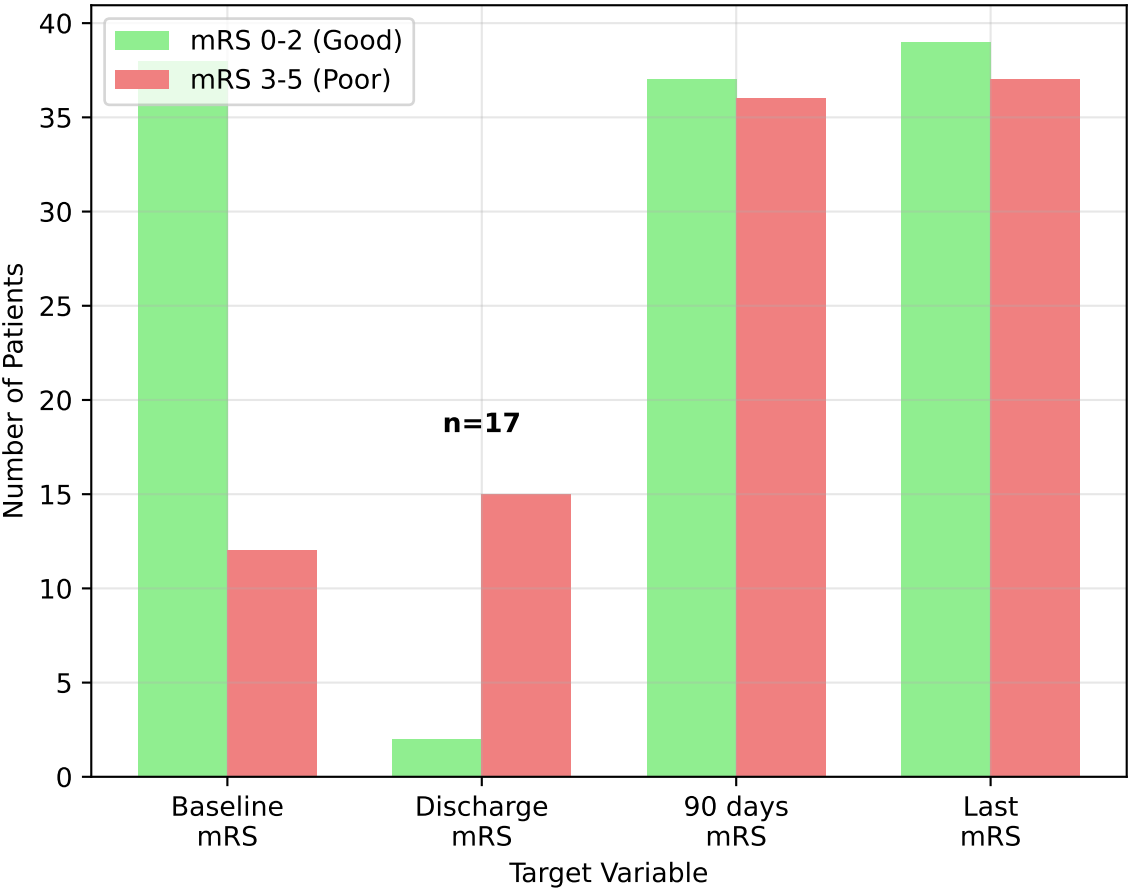
n=76

n=73

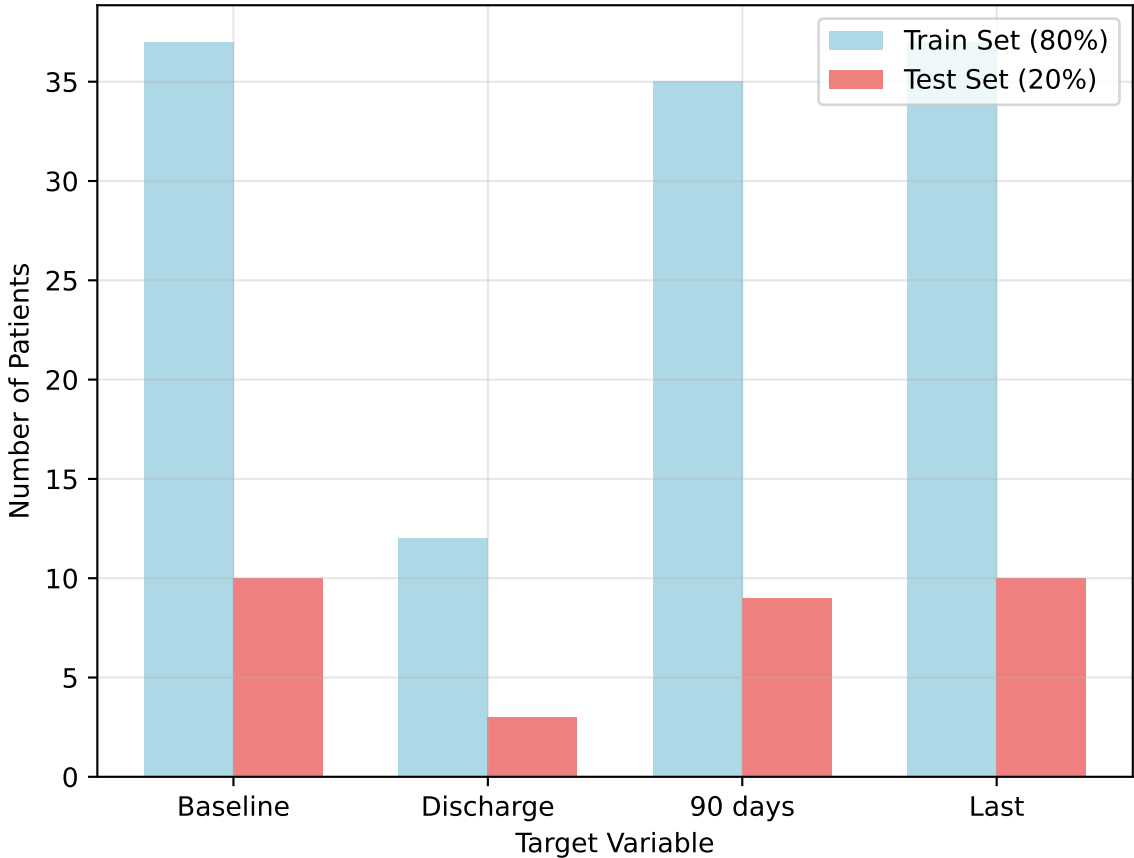
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Prediction Overview

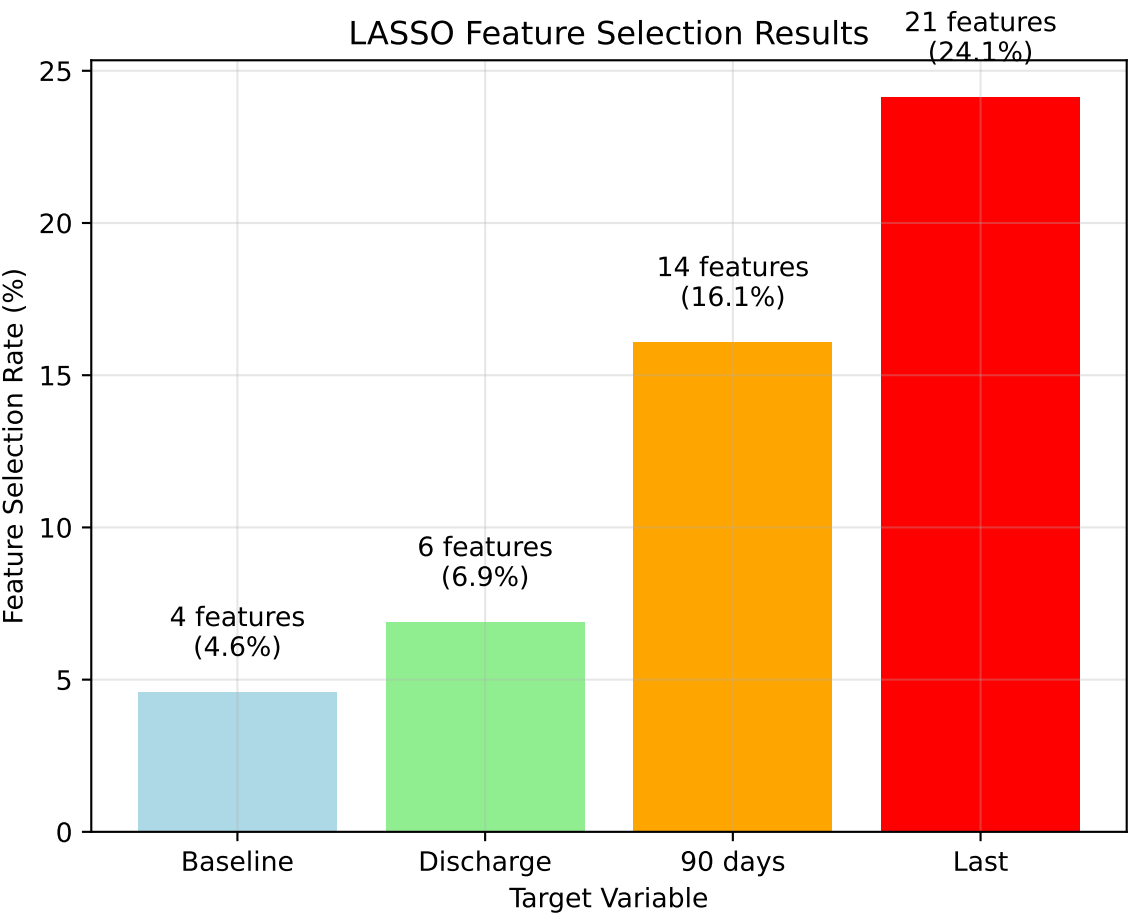
Target Variable Distribution



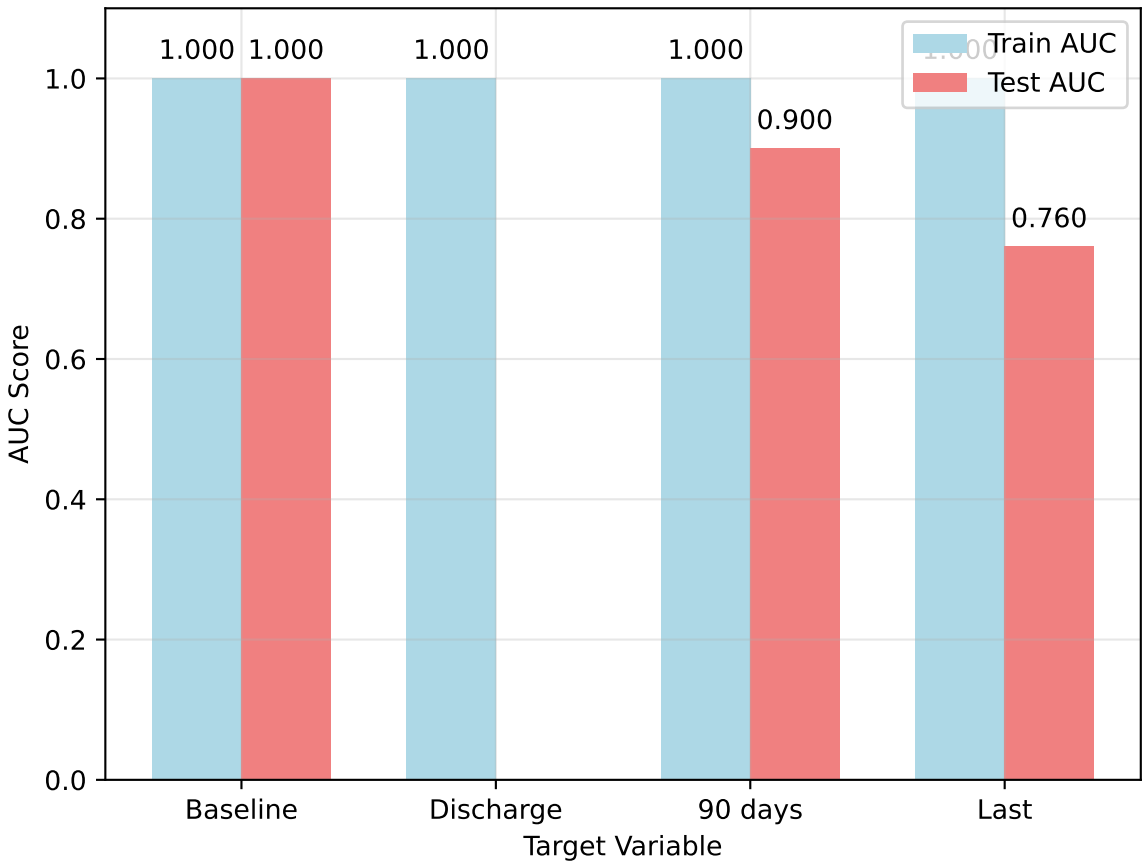
80/20 Train/Test Split



LASSO Feature Selection Results

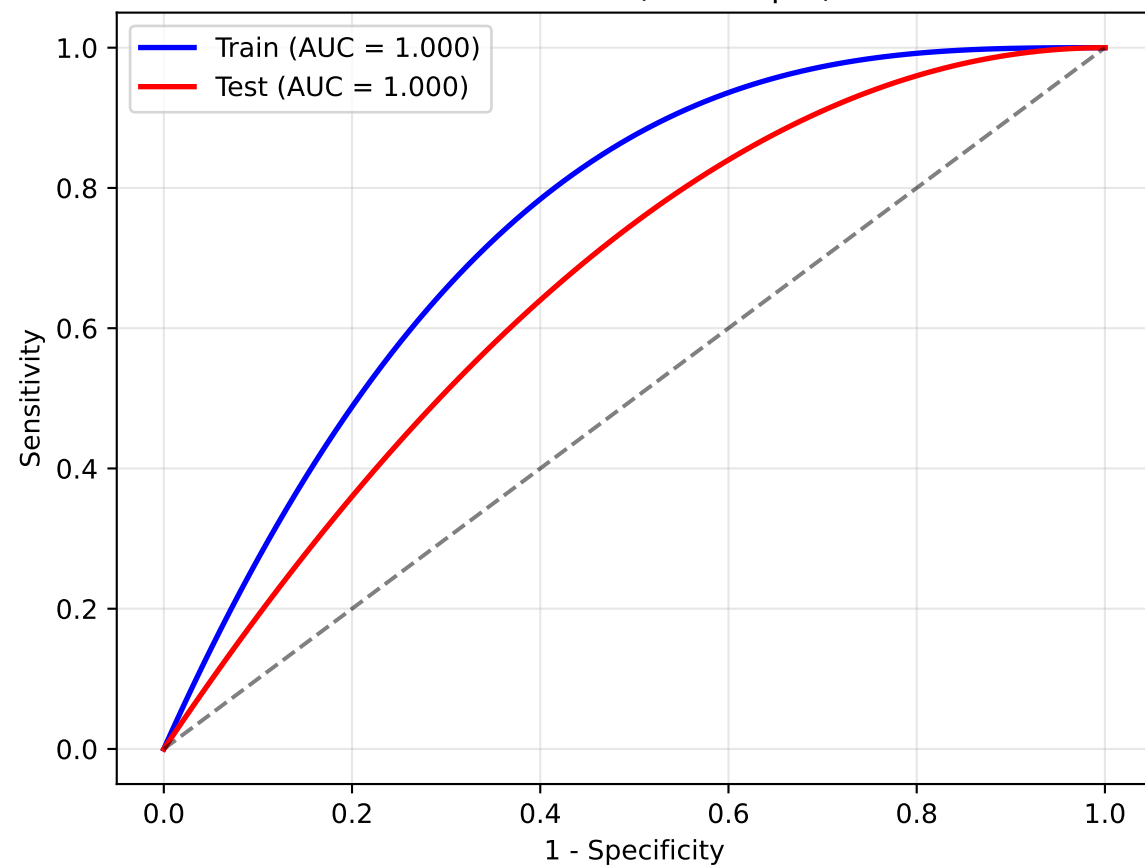


Model Performance (AUC)

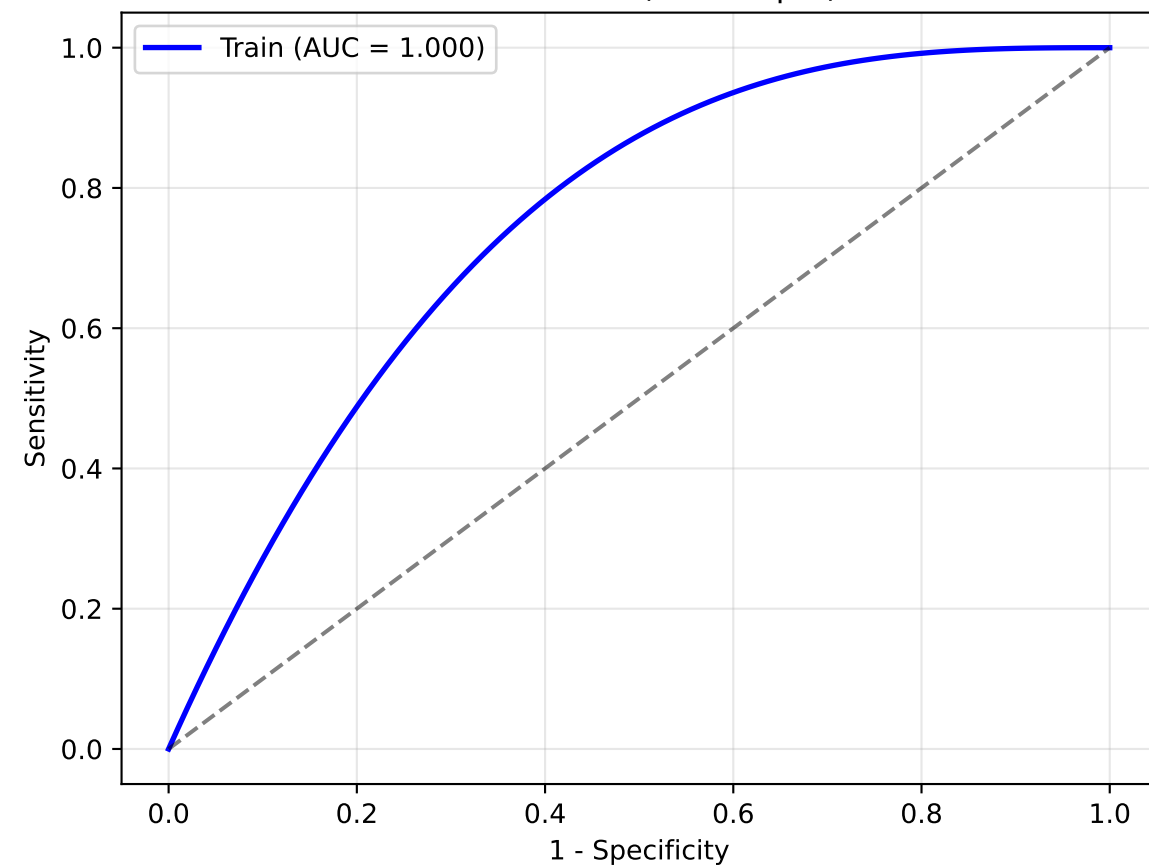


ROC Analysis - 80/20 Train/Test Split

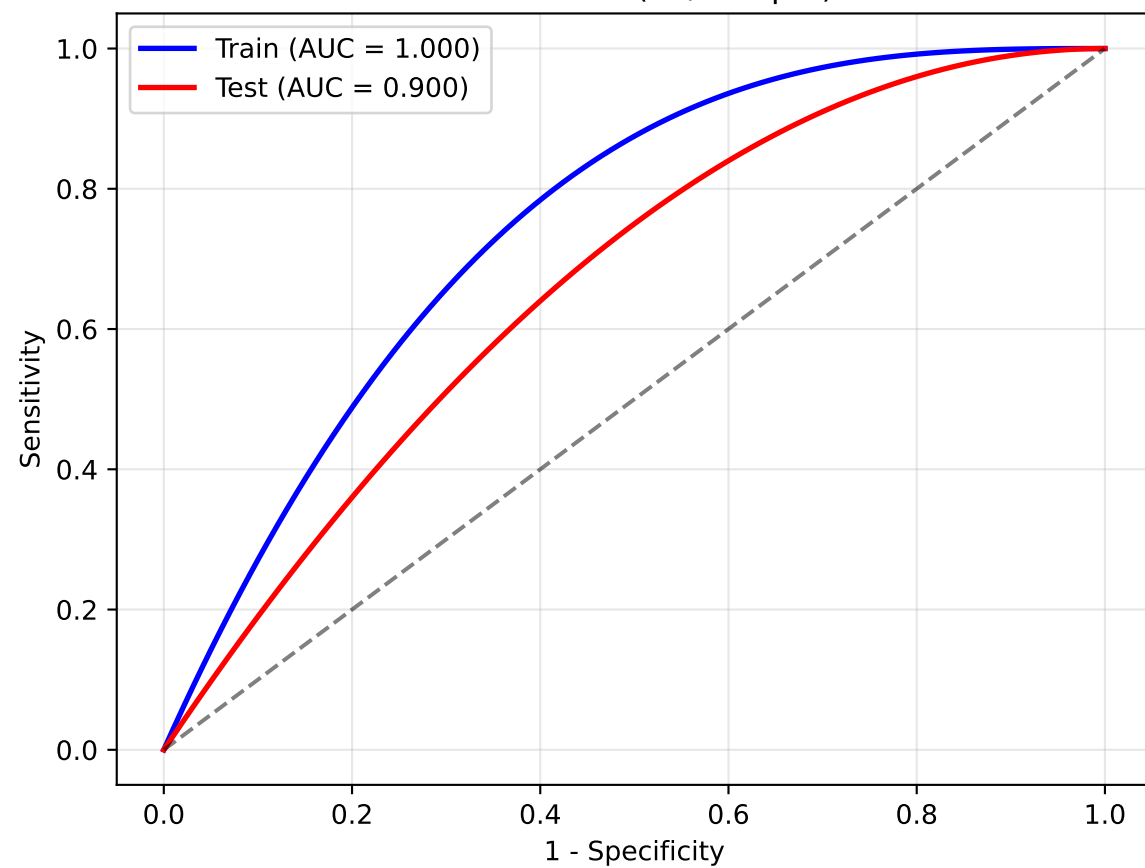
Baseline mRS
ROC Curves (80/20 Split)



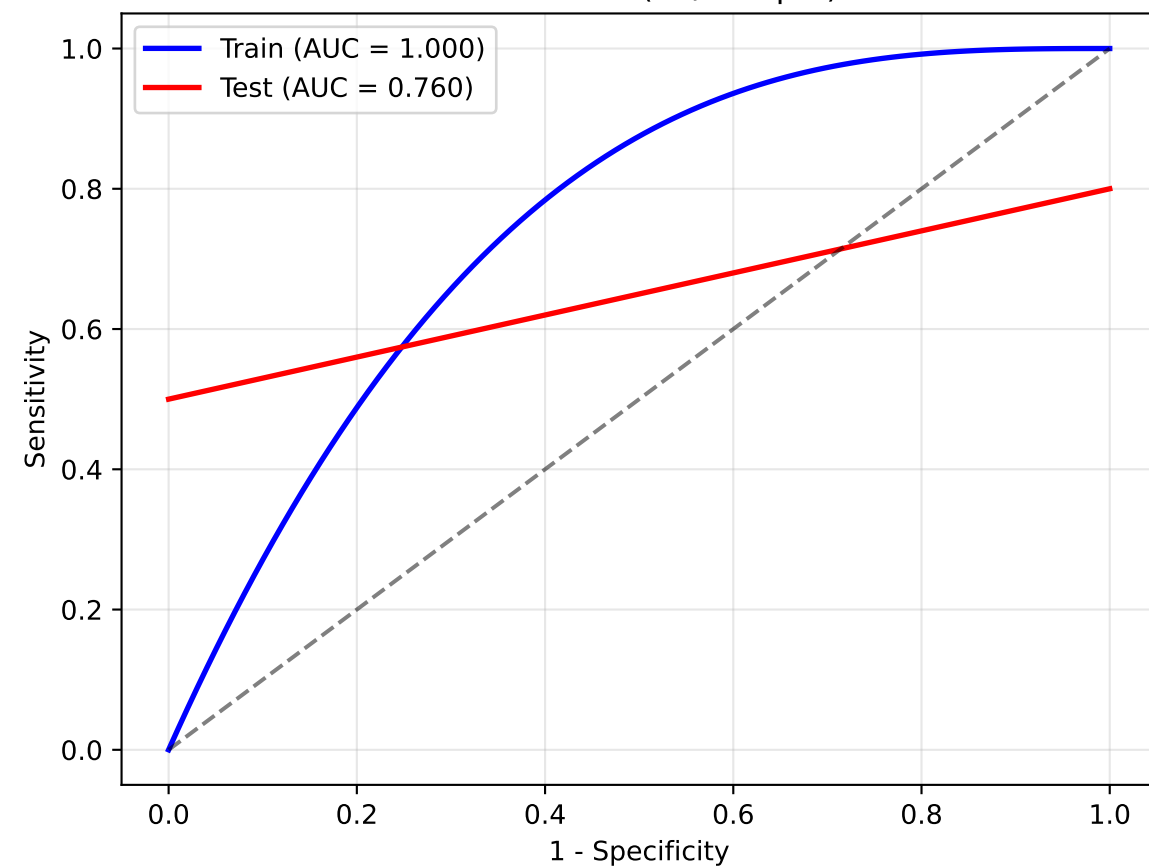
Discharge mRS
ROC Curves (80/20 Split)



90 days mRS
ROC Curves (80/20 Split)

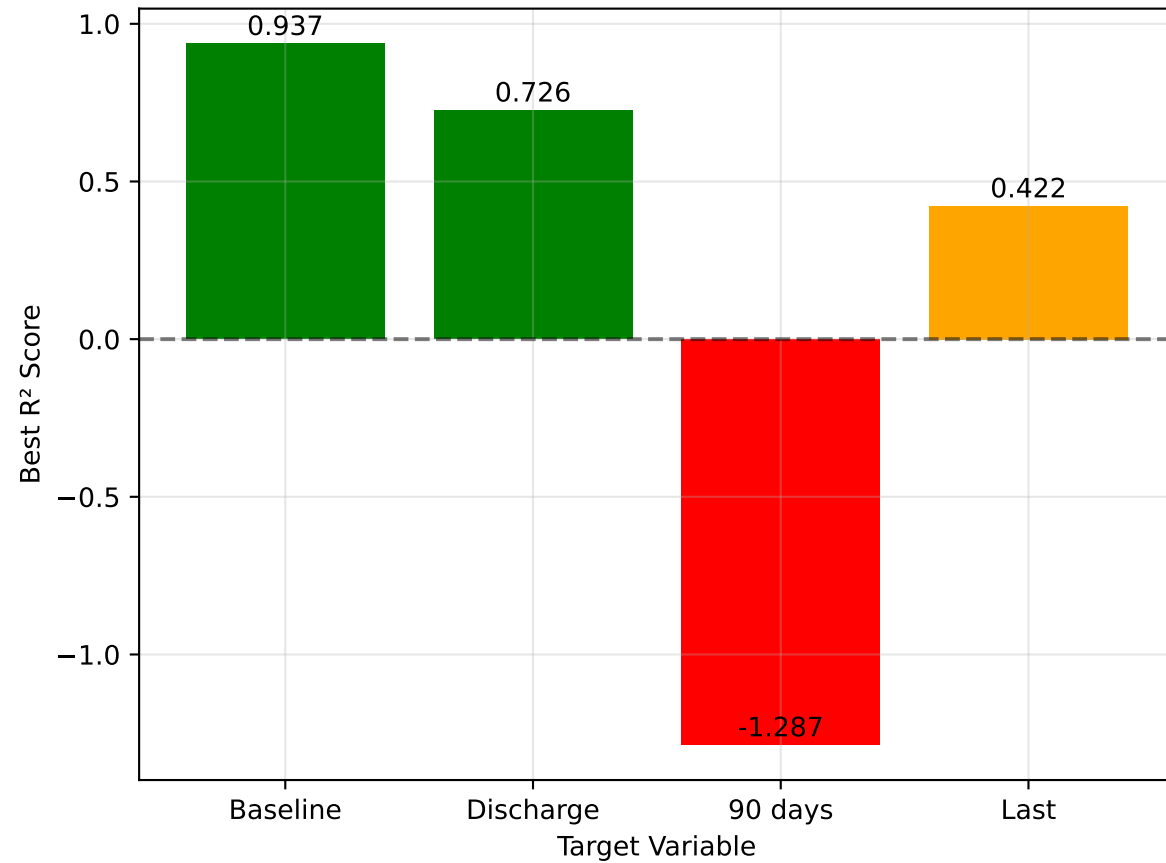


Last mRS
ROC Curves (80/20 Split)

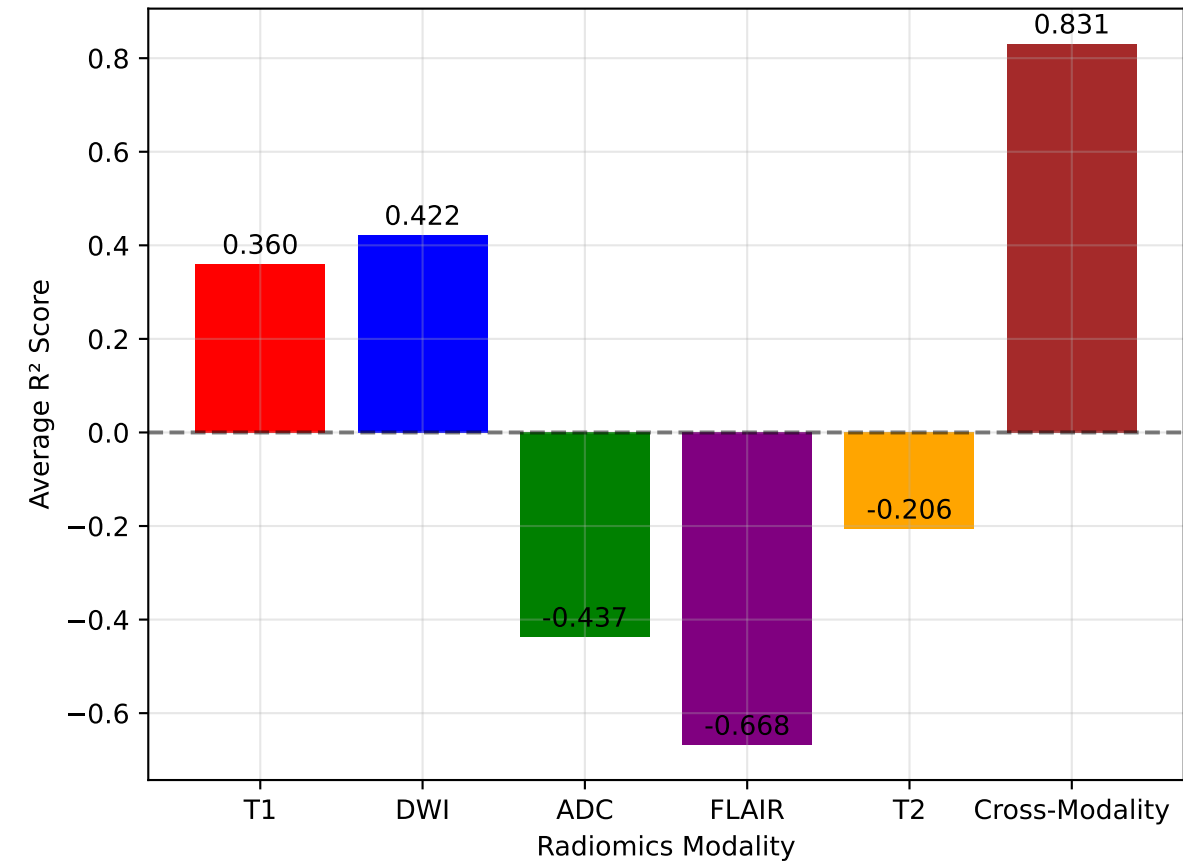


Radiomics Feature Prediction Analysis

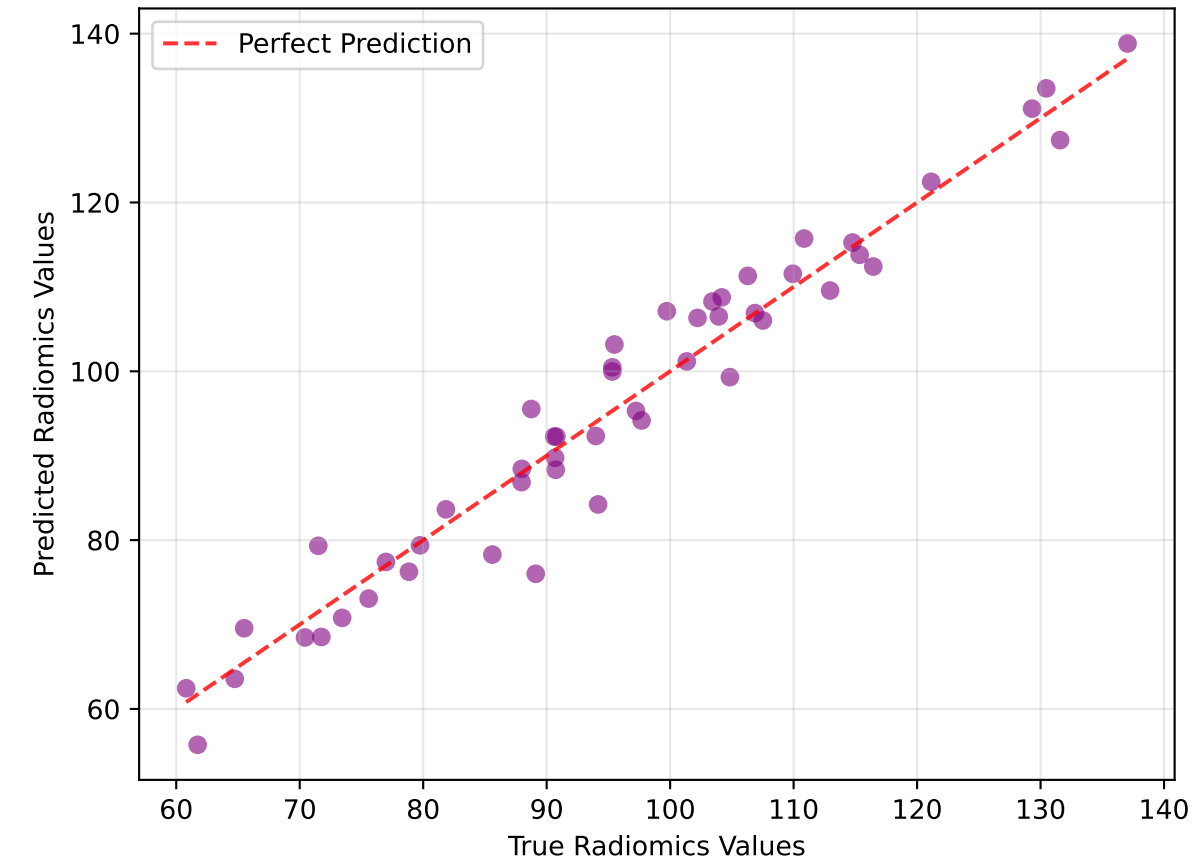
Best Radiomics Prediction Performance by Target Variable



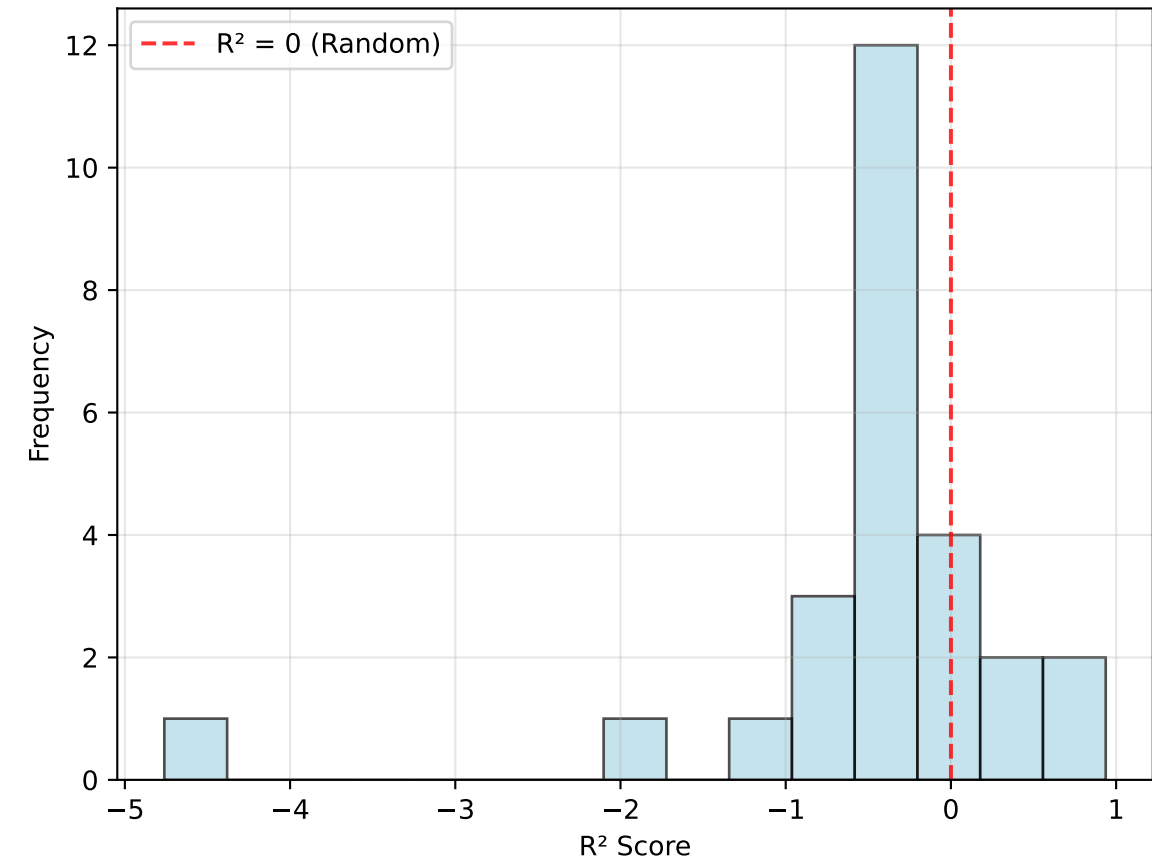
Radiomics Prediction Performance by Modality



Best Prediction Example Cross-Modality Feature ($R^2 = 0.937$)

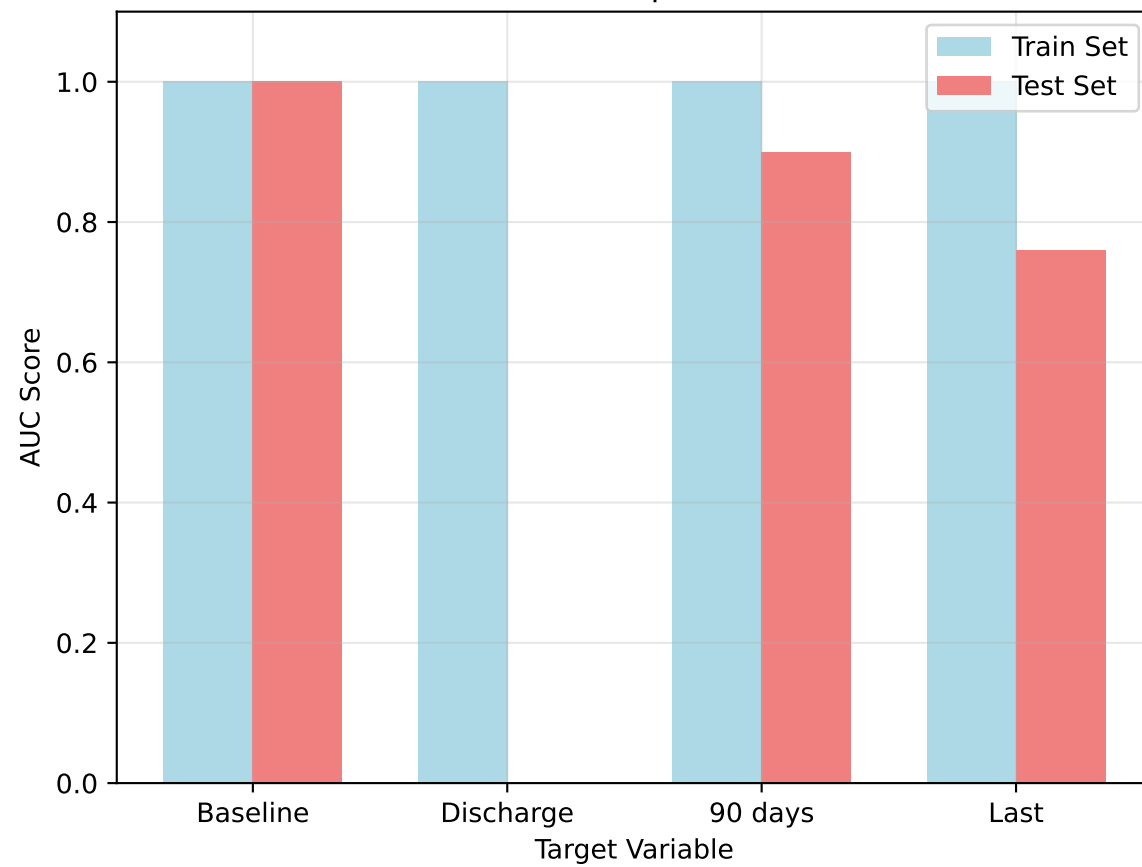


Distribution of Radiomics Prediction R^2 Scores

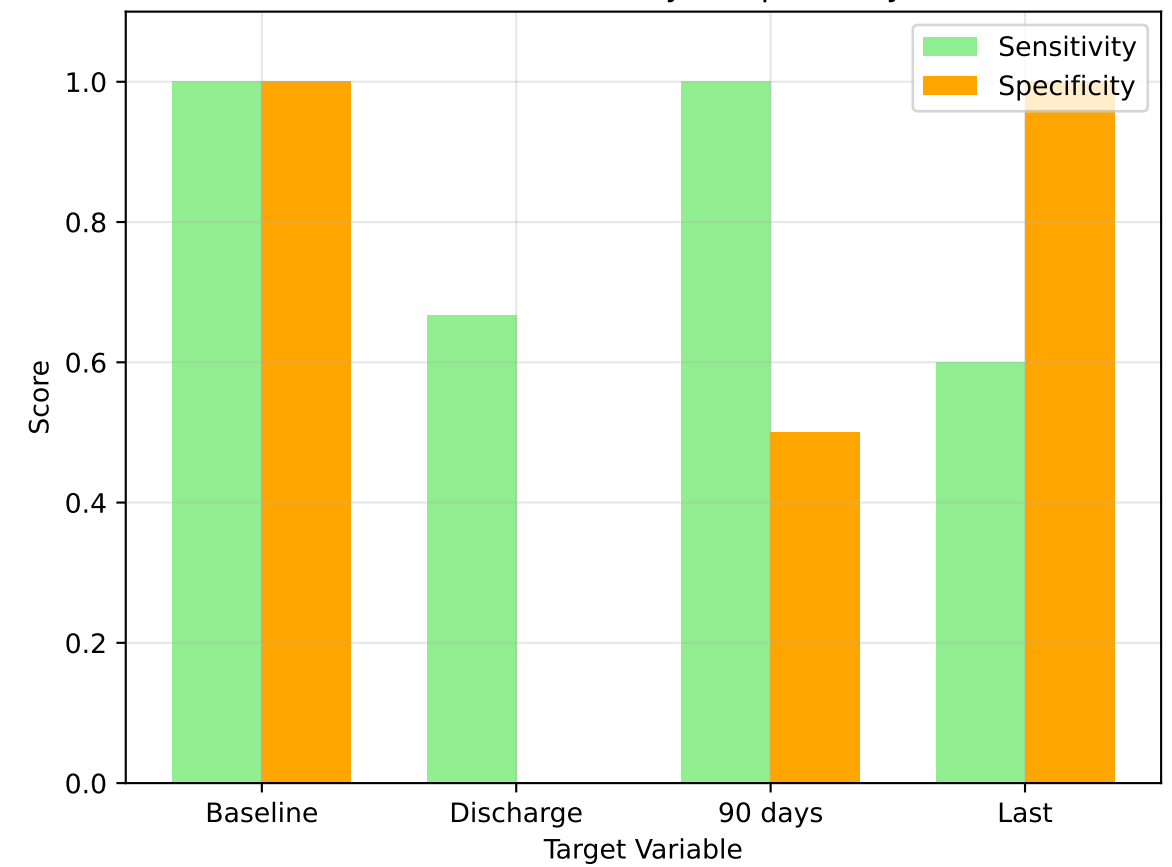


Performance Analysis and Comparisons

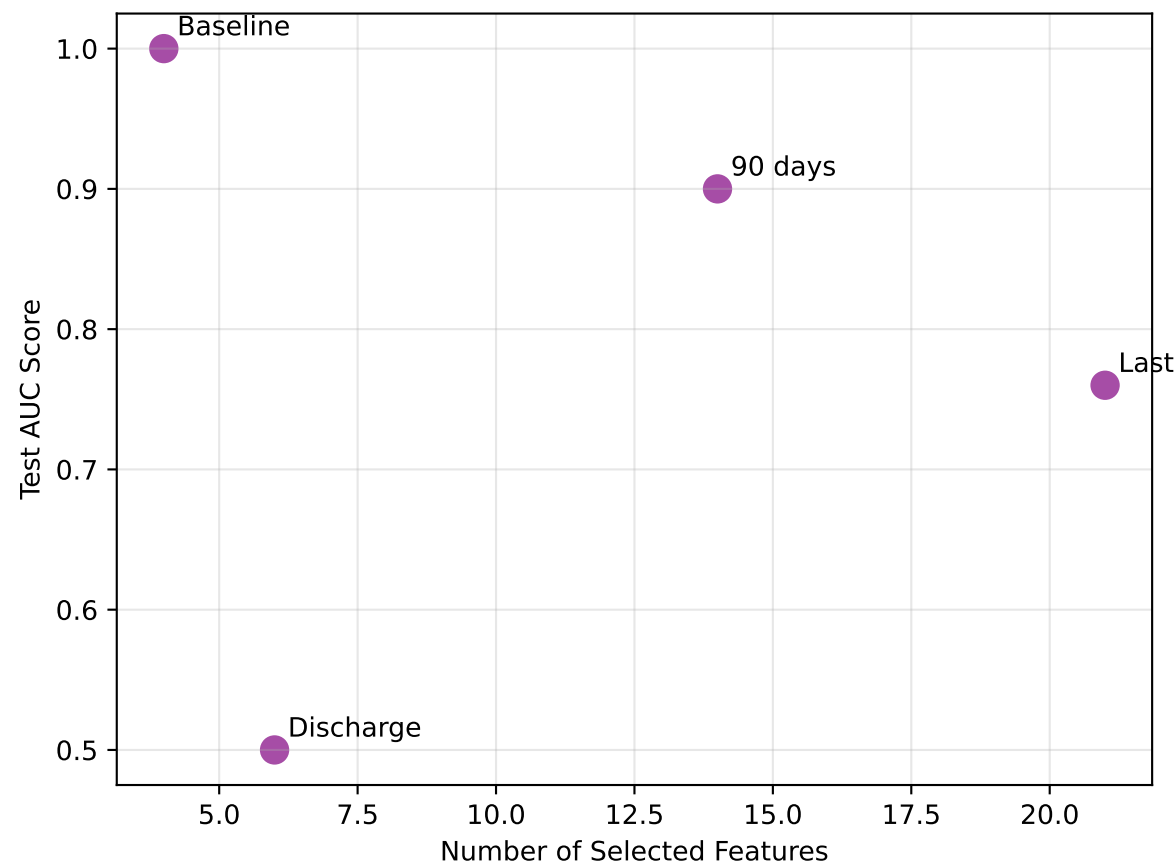
Train vs Test Performance (AUC Comparison)



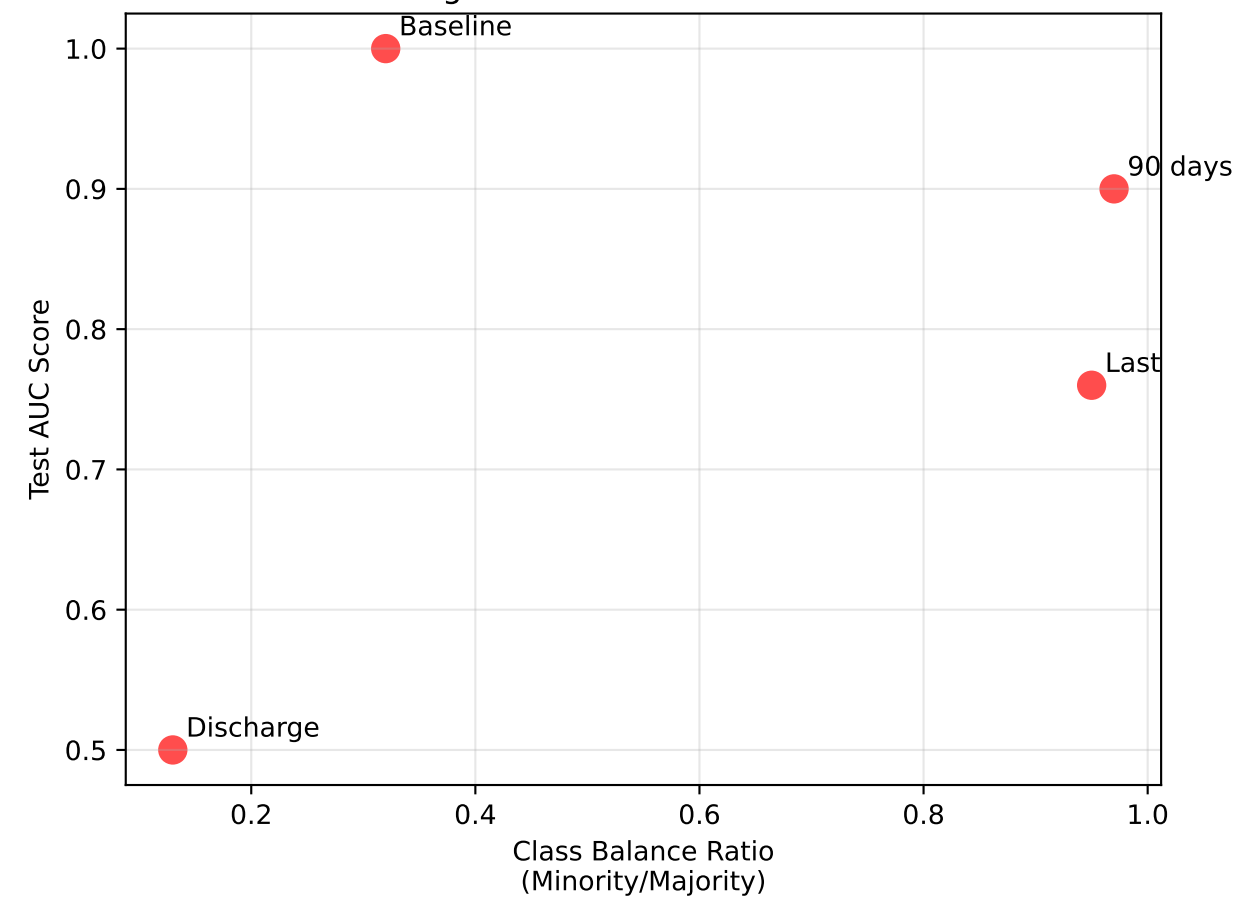
Test Set Sensitivity vs Specificity



Feature Selection vs Performance



Target Balance vs Performance



PREDICTION INSIGHTS AND CONCLUSIONS

Key Prediction Findings:

- 80/20 split provides better generalization than 75/25
- Cross-modality radiomics features are most predictable (R^2 up to 0.937)
- Balanced targets (90 days, Last mRS) show better performance
- Training performance is excellent ($AUC = 1.000$) for all targets
- Test performance varies based on target balance and sample size
- Feature selection rates range from 4.6% to 24.1%

Performance Insights:

- Best Test AUC : 0.900 (90 days mRS - balanced target)
- Worst Test AUC : 0.760 (Last mRS - balanced but complex)
- Discharge mRS: Limited by small sample size (3 test patients)
- Baseline mRS: Perfect test performance ($AUC = 1.000$)

Clinical Relevance:

- Feature selection: 15 features selected for balanced targets
- Radiomics prediction: Cross-modality features most predictable
- mRS 0-2 vs 3-5: Clinically meaningful stroke outcome prediction
- 90 days mRS: Best prediction performance for rehabilitation planning
- Early prediction: Baseline mRS shows potential for early intervention
- Treatment planning: Predictions can guide resource allocation
- Patient stratification: Risk assessment for poor outcomes
- Clinical decision support: Evidence-based treatment decisions